

Mohammad Hamzah Warsi

Electronics Engineer | Hardware Design, Embedded Firmware, Test & Automation

+971 50 944 8297 | hamzahw.work@gmail.com | linkedin.com/in/hamzahwarsi

Based in the UAE

PROFESSIONAL SUMMARY

Electronics engineer (MEng, Electrical & Electronic Engineering, University of Nottingham) with international experience in Belgium, Germany, Malaysia and Norway across embedded firmware, hardware design, testing, industrial automation and machine learning. At PEC NV in Belgium, developed C firmware on TI AM64x Arm processors, modified PCB designs and validated high-power battery cell test towers for BMW, Fraunhofer and Varta, including a rapid on-site intervention at BMW Wackersdorf that completed final testing for a €100M customer handover.

Strong in fault diagnosis and root-cause analysis, from a reversed valve behind a compressor overheating fault to incorrect wire selection behind a safety issue at a customer. Cut production downtime by 30% on a live Sony production line, and built machine learning forecasting models for hydropower and wind planning. First author of four research papers, two published (Springer Nature, IEEE) and two under review, on energy harvesting for highways and edge AI. A hands-on problem-solver who works well with customers and multidisciplinary teams.

CORE SKILLS

- **Hardware Design:** Analogue, digital and power electronics, component selection, schematic capture, PCB design and modification (Altium, KiCad, 4-layer), switch-mode power supplies, DC-DC converters, op-amp signal conditioning, power conditioning and battery charging circuits, LTspice
- **Embedded Firmware:** C, C++, TI AM64x (Arm Cortex-A53, Cortex-R5F, Cortex-M4F), STM32 (ARM Cortex-M), Arduino, RTOS, embedded Linux, bare-metal, state machines, cross-compilation, Keil uVision, I2C, SPI, CAN, UART, RS232, ADC, GPIO, Git, code reviews
- **Test & Validation:** Hardware-in-the-loop (HIL) testing, unit testing, functional validation, calibration, test plans and documentation, oscilloscope, multimeter, signal generator, bench power supply, thermal camera, SWD debugger, soldering, functional safety checks, root-cause analysis
- **Sensing & Instrumentation:** Pressure transmitters and transducers, thermal control and temperature sensing, Doppler radar signal processing, vacuum systems, ADC scaling, real-time data logging, HMI, alarm and fault-detection logic
- **Automation & Industrial:** Industrial robots (Fanuc, Mitsubishi), electrical cabinets and control panels, power systems, sensors and actuators (valves, pumps, fans, stepper motors), PID control
- **Machine Learning & Computer Vision:** Python, PyTorch, TensorFlow, Keras, TensorFlow Lite, 8-bit quantisation, scikit-learn, pandas, NumPy, OpenCV, YOLOV3 object and shape detection, edge deployment (Coral Edge TPU, GPU, TPU, VPU), time-series forecasting, data pipelines
- **Tools & Simulation:** MATLAB, Simulink, Ansys (mechanical and thermal), AutoCAD, 3D printing, Excel dashboards, MS Office, AI-assisted development tools

PROFESSIONAL EXPERIENCE

Electronics Engineer, Firmware & Testing | Feb 2025 – Dec 2025

PEC NV, Belgium

Integrated cell tester towers for BMW, Fraunhofer and Varta: firmware, safety, calibration, instrumentation and customer support.

Firmware & Software Development

- Reduced test-cycle variability on automated battery test systems for BMW and Fraunhofer by developing C firmware across embedded Linux, RTOS and bare-metal on TI AM64x Arm processors, using cross-compilation, Git version control and code reviews.

- Added new docking-controller functions to the tester firmware and wrote the control logic as state machines.
- Validated software on real equipment in hardware-in-the-loop (HIL) tester setups, and unit tested the new cell pallet.
- Debugged communication and signal faults across I2C, SPI, CAN, UART and RS232 using an SWD debugger and oscilloscope, working with hardware and firmware engineers.
- Sped up scripting, development and debugging by using AI-assisted coding tools.

AI8 Analogue Input Cards

- Improved analogue input accuracy, proven by documented before/after performance metrics, by developing a firmware calibration update for AI8 (Analog Input 8) modules in Keil uVision and refining the calibration lookup tables and gain offsets.
- Validated every change with functional testing before release.

Safety System

- Made faults faster to diagnose by reformatting the alarm system for service engineers, and introduced new alarms for bimetal flicker and air-pump failure.
- Made the system fail safe with a digital-output health check that cross-validates outputs against inputs, so the system becomes inoperable until the fault is resolved.
- Protected operators by implementing evaporator-fan post-run logic that keeps the fan running for a set time after an overheat or at high temperatures, and functional safety checks for tester door seals.
- Improved system readiness by detecting the availability of the air pump, and prevented overheating by monitoring refrigerant leaks over a series of months.

Thermal Control Cards (THC)

- Brought 40 thermal control cards back to specification by calibrating, testing and validating each card for temperature readings.
- Fixed firmware flaws, including a sensor type misconfigured in the registers and incorrect offset calculations.
- Reduced calibration time and manual steps by modifying and automating the calibration scripts, and reported accuracy across all calibrations in an Excel dashboard.

Vacuum System & Pressure Measurement

- Implemented a vacuum system into the testing stations.
- Enabled real-time pressure monitoring from -1 to 12 bar by implementing pressure transmitter functionality, with ADC channel configuration and scaling algorithms that achieved $\pm 1\%$ measurement accuracy.
- Validated performance against the transmitter datasheet specifications, and integrated live pressure data into the HMI for operator visibility, data logging and overpressure safety alerts.

Thermal Management & Compressor Fault (Cross-Department Project)

- Added new functions to the cooling system in a cross-department project by implementing 3-way-valve code for flow direction and regulation and fan control for thermal management, verified through functional testing.
- Found the cause of a compressor overheating fault that the firmware could not explain by placing temperature sensors on the valve pipes and comparing readings across test stages against a working unit. This revealed a stepper-motor valve fitted in reverse, which could not be fixed in-house and was repaired externally.

Type C Pallet & Prototyping

- Reduced contact-resistance variability by identifying and resolving key design flaws in the next-generation Type C pallet, now deployed at BMW for battery cell testing.
- Implemented different probes and cell treatments for maximum efficiency in testing cells.

- Took a new design from prototype toward production by prototyping and unit testing a 20-cell pallet for BMW "Neue Klasse" cells.
- Improved existing hardware by modifying PCB designs, and contributed to informal design risk reviews that flagged failure modes such as overheating and overpressure.
- Developed comprehensive test documentation and best-practice procedures for minimum contact resistance.

BMW Wackersdorf, Germany (1-Week Rapid Intervention)

- Resolved critical tower issues on-site by troubleshooting and providing solutions across the assigned towers.
- Eliminated a recurring overheating risk by diagnosing and fixing control-panel wiring faults.
- Improved measurement reliability by correcting sensor placement and design flaws.
- Identified the root cause of a condensation issue and wrote testing documentation for future service engineers.
- Successfully completed final testing of tower RD01 and delivered it to BMW, contributing to a €100M customer handover, one of the company's biggest.

Customer Projects: Fraunhofer and Varta

- Gave Fraunhofer a clear, data-backed recommendation by designing and executing a comparative study of 2 new contact probes across 3 surface treatments (untreated, laser-treated, sandpaper) and multiple power inputs, measuring contact resistance and maximum temperature at the contact points.
- Identified Probe X as superior across most test conditions, documented all results, delivered a final report with a recommendation, and estimated the manufacturing costs of the options.
- Resolved a critical safety issue for Varta, where the customer's contact probes exceeded temperature limits, by running controlled tests with the customer's test regime across multiple wire types and identifying incorrect wire selection as the root cause.
- Restored normal operation by recommending replacement of the wires, and documented findings and corrective actions for the customer's quality team.

Automation Engineer Intern | Jun 2022 – Sep 2022

Sony EMCS, Malaysia

- Cut production downtime by 30% within 3 months by deploying and fine-tuning Fanuc and Mitsubishi robots on a live TV production line.
- Kept production running by leading control panel and power system troubleshooting, and supported electrical cabinet and control panel work for the robot cells.
- Sped up root-cause analysis by designing an Excel dashboard that tracked every robot fault and stoppage.

Machine Learning Engineer | Feb 2022 – May 2022

Think Outside, Norway

- Enabled 5-year energy planning for hydropower and wind integration by developing time-series forecasting models for electricity prices and water availability in Python, using scikit-learn, pandas and NumPy.
- Improved forecast accuracy by owning the end-to-end data pipeline: data cleaning, preprocessing, feature selection and model tuning.
- Made results reproducible and comparable by establishing machine learning workflows with data versioning and experiment tracking.

ENGINEERING PROJECTS

- **Energy Harvesting System for Highways:** Designed a multi-source system (piezoelectric, thermoelectric, wind) that captures waste heat, vibration and airflow to power street lighting and utilities, projected at 350W at highway speeds. Engineered the power conditioning and 12V battery charging circuit, including a boost converter that holds a 10–20V input at 24V with under 2 mV ripple, and modelled the system in ANSYS (mechanical and thermal), LTspice and Simulink.
- **Edge-Based Coffee Leaf Disease Classification:** Trained five deep learning models (PentaCNN, MobileNetV2, VGG-16, VGG-19, DenseNet121) in Python on the JmuBen dataset using GPU and TPU, deployed the best model to GPU, TPU and VPU, and built a benchmarking solution comparing all three platforms. With TensorFlow Lite and 8-bit quantisation, the Coral Edge TPU reached 10 ms inference against 40 ms on a GPU workstation, and MobileNetV2 reached 98.9% training accuracy (first-author IEEE paper).
- **Custom Switch-Mode Power Supply:** Engineered a 12V/2A supply (TL494 controller, 100 kHz) incorporating a transformer, supercapacitor bank, inverters and relay control coils, with analytical calculations of component ratings, efficiency and thermal performance. Designed and routed a 4-layer PCB in KiCad, and reached ~85% efficiency with 150 mV ripple.
- **Face and Shape Detection (OpenCV, YOLOv3):** Trained a custom detection model from YOLOv3 weights on a large self-built dataset of face images, augmented for lighting, angle and expression to improve robustness. Implemented shape detection for circles, squares and triangles, with real-time detection in OpenCV and Python.
- **Doppler Radar Speed Gun:** Built a speed gun from a Doppler radar module (HB100) and an STM32 microcontroller, with op-amp signal conditioning and a signal processing algorithm that converts the Doppler frequency shift to velocity using zero-crossing detection. Measured speed within $\pm 5\%$ of a commercial radar gun and displayed it in real time on an LED interface.
- **Line-Following 4WD Hercules Robot:** Designed and etched copper PCBs for an 8-sensor infrared array, wrote Arduino firmware for sensor processing and PID motor control, and modelled and 3D-printed a custom IR sensor holder in AutoCAD, achieving stable line tracking on the test course.

EDUCATION

MEng, Electrical & Electronic Engineering | 2019 – 2024

University of Nottingham

Foundation in Engineering | 2017 – 2018

University of Nottingham

CERTIFICATIONS

- C Language, Basic to Advanced | Learnvern | 2026
- Legacy JavaScript Algorithms and Data Structures | freeCodeCamp | May 2023
- Data Analysis with Python | freeCodeCamp | Sep 2022
- Machine Learning with Python | freeCodeCamp | Aug 2022
- Responsive Web Design | freeCodeCamp | Aug 2022
- Big Data 101 | IBM | Oct 2021
- Data Visualisation using Python | IBM | Oct 2021
- Python for Pentesters | Alison | Oct 2021
- Computer Networking – Wired and Wireless | Alison | Sep 2021

PUBLICATIONS

- **Advances in Energy Harnessing Techniques for Smart Highways: A Review** | First author | Springer Nature, Electrical Engineering, 106(5), 6389–6408 | Published 14 Apr 2024 | DOI: [10.1007/s00202-024-02379-8](https://doi.org/10.1007/s00202-024-02379-8). Reviews piezoelectric, thermoelectric, pyroelectric, electromagnetic and solar highway energy harvesting, with their principles, challenges and trade-offs.
- **Real-Time Disease Classification in Coffee Plants with Coral Developer Board Mini** | First author | IEEE International Conference on Agrosystem Engineering, Technology & Applications (AGRETA) | Published 9 Jul 2024 | DOI: [10.1109/AGRETA61912.2024.10948998](https://doi.org/10.1109/AGRETA61912.2024.10948998). Shows an automated coffee leaf disease classifier on the Coral Developer Board Mini with Google's Edge TPU, where MobileNetV2 reached 98.9% training accuracy and the Coral board ran inference in 10 ms against 40 ms on a GPU workstation.
- **Detecting Coffee Leaf Diseases with the Google Coral Developer Board Mini** | First author | Computers and Electronics in Agriculture | Under revision (first correction)
- **Wired Highways: Energy Production for Roads** | First author | Open Access Journal of Power and Energy (OAJPE) | Submitted after first correction

SPORT & ACTIVITIES

Represented national side in football and swimming at the University of Nottingham Nations Cup | Football for school and university teams, including inter-school competitions | Swimming for school in amateur competitions

LANGUAGES

English (Professional, spoken and written) | Hindi (Native) | Urdu (Native) | Malay (Intermediate, spoken and written)